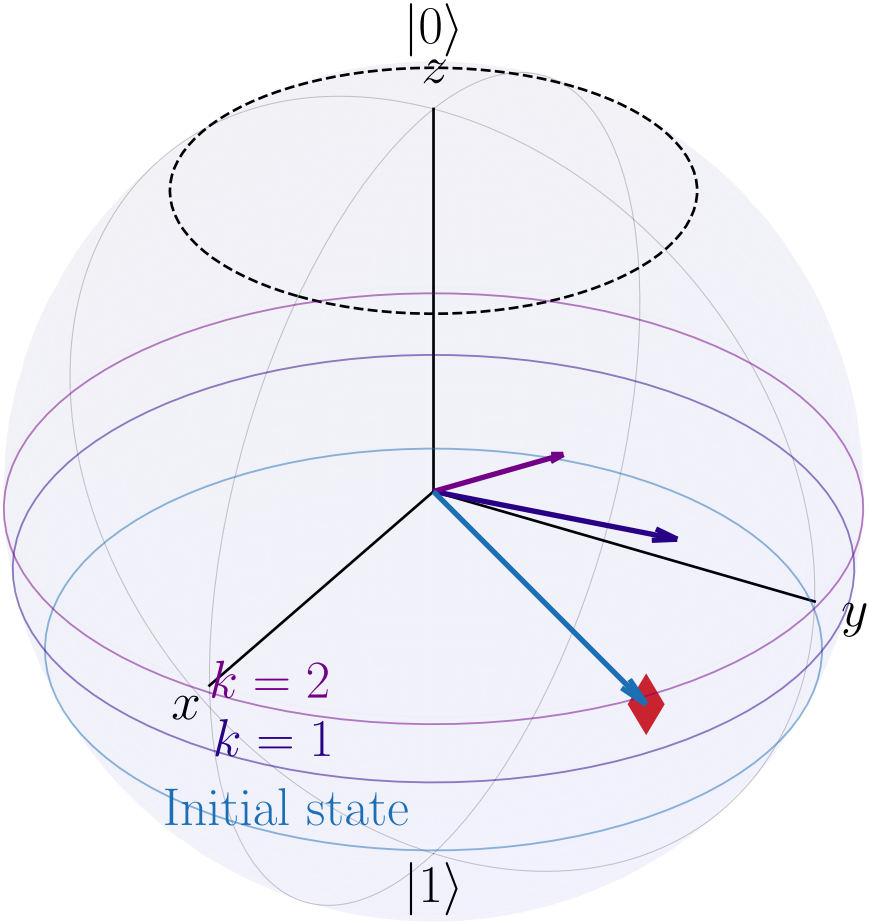

$|0\rangle$
$z$
$y$
$x$
$k = 2$
$k = 1$
Initial state
$|1\rangle$